

TrES-1b

R-1346

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_190528 · created 2026-07-31T22:07:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458631.8416 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1355 +/- 0.0081
Transit depth (Rp/Rs)^2	1.83 +/- 0.22 [%]
Orbital Inclination (inc)	88.33 +/- 0.77
Airmass coefficient 1 (a1)	1.0866 +/- 0.0096
Airmass coefficient 2 (a2)	-0.0662 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	0.89 %
Best Comparison Star	#2 - [46, 42]
Optimal Aperture	5.05
Optimal Annulus	12.63
Transit Duration (day)	0.0995 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1355 ± 0.0081 vs 0.1358 (deviation 0.04σ)
Duration fit vs published	0.0995 d vs 0.1042 d (deviation 1.09σ)
Mid-time O–C	-1.6 min
Depth SNR	16.7
Residual scatter	0.89 %

Light curve

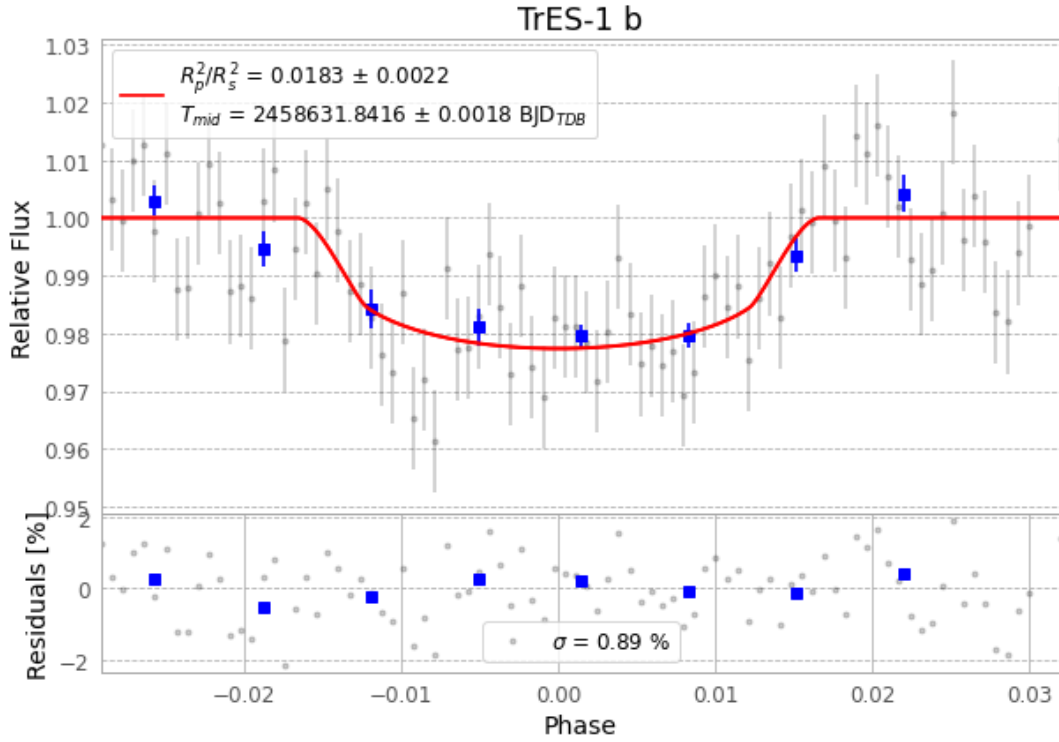


Figure 1 — FinalLightCurve_TrES-1 b_2019-05-27.png (SHA-256 5f442b6b2e1e2f27...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 96], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fdfb95adb7289ad4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1190528060012.FITS → TRES-1190528103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-190528.log (44dd67bf82ae...), FinalLightCurve_TrES-1 b_2019-05-27.png (5f442b6b2e1e...), FinalLightCurve_TrES-1 b_2019-05-27.csv (4a684312af60...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 22:07Z.